GATE DA Notes — Maths, Programming and Databases

Probability and Statistics

- **Counting:** P(n,r) = n!/(n−r)!; C(n,r) = n!/(r!(n−r)!). Bijections of {1..n} that satisfy f(f(x)) = x are the involutions: 1, 2, 4, 10 for n = 1..4.
- **Conditional probability:** $P(A|B) = P(A \cap B)/P(B)$. **Bayes:** $P(A|B) = P(B|A)P(A)/P(B)$. Total probability: $P(B) = \Sigma P(B|A_i)P(A_i)$.
- **Expectation and variance:** E[aX+b] = aE[X]+b; $Var(aX+b) = a^2 Var(X)$; $Var(X) = E[X^2] - (E[X])^2$. For independent X, Y: Var(X+Y) = Var X + Var Y. Cov(X,Y) = E[XY] − E[X]E[Y]; independence implies zero covariance, not conversely.
- **Law of total expectation:** E[X] = E[E[X|Y]]; Var(X) = E[Var(X|Y)] + Var(E[X|Y]).

Distribution	Mean	Variance
Bernoulli(p)	p	p(1−p)
Binomial(n,p)	np	np(1−p)
Poisson(λ)	λ	λ
Uniform(a,b)	(a+b)/2	$(b-a)^2/12$
Exponential (rate λ)	1/λ	$1/\lambda^2$
Normal(μ,σ²)	μ	σ^2

- **Exponential:** P(X > x) = e^(−λx); memoryless. (If a question gives the *mean*, the rate is its reciprocal.)
- **Normal:** z = (x−μ)/σ; P(|Z|<1) ≈ 0.68, P(|Z|<2) ≈ 0.95.
- **CLT:** the mean of n i.i.d. variables is approximately Normal(μ, σ²/n).
- **Confidence interval (known σ):** $\bar{x} \pm z \cdot \sigma/\sqrt{n}$ (z = 1.96 for 95%). With unknown σ use the t-distribution (n−1 degrees of freedom). Tests: z-test (known σ), t-test (unknown σ), χ^2 test (goodness of fit/independence).
- **Sample mean update:** new mean = (n·old + x)/(n+1).

Linear Algebra

- **Rank–nullity:** rank(A) + nullity(A) = number of columns. A system Ax = b is consistent iff rank(A) = rank([A|b]).
- **Eigenvalues:** Σλ = trace, Πλ = determinant; eigenvalues of A^k are λ^k; of A^{-1} are 1/λ. Triangular matrix: eigenvalues are the diagonal. Symmetric matrices have real eigenvalues and orthogonal eigenvectors. Rotation matrices (angle θ) have eigenvalues e^(±iθ).
- **Projection matrix** $P = A(A^TA)^{-1}A^T$: $P^2 = P$, symmetric, eigenvalues 0 or 1, trace = rank. The centring matrix $I - (1/n)\mathbf{1}\mathbf{1}^T$ is a projection of rank n−1 (eigenvalues 1 with multiplicity n−1, and 0).
- **Orthogonal matrix:** $Q^TQ = I$, |det Q| = 1, preserves lengths. **Idempotent:** $A^2 = A$.
- **Subspace test:** contains 0, closed under addition and scalar multiplication. A plane

through the origin in $\mathbb{R}^3$ is a 2-dimensional subspace; its intersection with a ball of radius r is a disc of area πr^2.
- **SVD:** $A = U\Sigma V^T$; singular values are square roots of eigenvalues of A^TA; rank = number of non-zero singular values; $\|A\|_F^2 = \Sigma\sigma_i^2$.
- **LU decomposition:** $A = LU$ by Gaussian elimination (without row swaps).
- **Quadratic form** x^TAx is positive definite iff all eigenvalues > 0.

Calculus and Optimisation

- **Continuity** at a: left limit = right limit = f(a). **Differentiable:** left and right derivatives agree (continuity is necessary, not sufficient). Piecewise functions: match values and slopes at the joins.
- **Maxima/minima:** set $f'(x) = 0$; $f'' > 0$ gives a local minimum, $f'' < 0$ a local maximum. On a closed interval also check the end points. On a half-open interval a maximum or minimum may not exist.
- **Taylor series:** $f(x) \approx f(a) + f'(a)(x-a) + f''(a)(x-a)^2/2 + \ldots$
- **Limits:** use L'Hôpital for 0/0 or ∞/∞. Standard: $\sin x / x \to 1$, $(1+1/n)^n \to e$.
- **Series:** geometric $\Sigma r^k = 1/(1-r)$ for $|r| < 1$; double sums of products factor into the product of sums.
- **Gradient descent:** $x \leftarrow x - \eta f'(x)$; for $f = \frac{1}{2}ax^2$, it converges iff $0 < \eta < 2/a$.

Programming, Data Structures and Algorithms

- **Python traps:** a mutable default argument (`def f(x, lst=[])`) is created once and shared across calls; slicing copies, assignment aliases; integer division `//`; recursion needs a base case; trace variables on paper.
- **Stack** (LIFO), **queue** (FIFO), **deque** (insert/remove at both ends); linked list: O(1) insert at head, O(n) search; **hash table:** O(1) average lookup, chaining or open addressing; expected probes for unsuccessful search with uniform hashing $\approx 1/(1-\alpha)$.
- **Binary tree:** with n leaves in a full binary tree there are n−1 internal nodes. A tree is determined uniquely by pre-order plus in-order (or post-order plus in-order); pre-order plus post-order does not fix it in general.
- **Searching:** linear O(n); binary search O(log n) on a sorted array; maximum comparisons $\approx \lfloor \log_2 n \rfloor + 1$ (so 10 for n = 1000).
- **Sorting:**

Algorithm	Best	Average	Worst	Stable
Selection	n^2	n^2	n^2	no
Bubble (with early exit)	n	n^2	n^2	yes
Insertion	n	n^2	n^2	yes
Merge	n log n	n log n	n log n	yes
Quick	n log n	n log n	n^2	no

- **Divide and conquer:** merge sort $T(n) = 2T(n/2) + n$; quicksort worst case on already-

sorted input with first/last pivot is n^2.

- **Graphs:** BFS (queue) gives shortest paths in unweighted graphs; DFS (stack/recursion) gives pre/post order and topological sort of a DAG (reverse post-order); Dijkstra for non-negative weights, O((V+E) log V) with a heap; Bellman–Ford for negative edges.

Databases and Warehousing

- **Relational algebra:** σ (select), π (project), ⋈ (join), ∪, ∩, −, × (product), ρ (rename). Left outer join keeps unmatched left rows with NULLs.
- **SQL:** `GROUP BY` with aggregates; `HAVING` filters groups, `WHERE` filters rows; `COUNT(*)` counts rows, `COUNT(col)` skips NULLs; `NOT IN` with a NULL in the subquery returns no rows.
- **Functional dependencies:** closure X^+ gives candidate keys; a key's closure is all attributes. **BCNF:** every non-trivial FD X→Y has X as a superkey. **3NF:** also allows Y to be a prime attribute. 2NF removes partial dependencies on a key.
- **ER model:** a many-to-many relationship becomes its own table; a one-to-many relationship adds a foreign key on the "many" side; weak entities need the owner's key.
- **B+ tree:** all data in leaves, leaves linked; order p means an internal node has at most p pointers (p−1 keys). Height ≈ log over the fan-out; splitting a full leaf promotes the middle key.
- **Indexing:** dense vs sparse; clustered vs non-clustered; a search through a B+ tree costs height + 1 block reads.
- **Data transformation:** min-max scaling, z-score, discretisation (equal-width, equal-frequency), sampling, compression.
- **Data warehouse:** star schema (one fact table, denormalised dimensions); snowflake (normalised dimensions). **Concept hierarchy:** e.g. day → month → quarter → year. Cuboids: with n dimensions and no hierarchy there are 2^n cuboids; with hierarchy levels L_i per dimension, the total is $\Pi(L_i + 1)$. OLAP: roll-up, drill-down, slice, dice, pivot.

Revision checklist

- ☐ Bayes and total probability by hand; exponential/Poisson formulas
- ☐ Eigenvalue tricks (trace, determinant, powers); projection matrices
- ☐ Rank–nullity; SVD facts
- ☐ Piecewise continuity and differentiability
- ☐ Python tracing: default arguments, recursion
- ☐ Complexity table for sorting; BFS/DFS/Dijkstra
- ☐ Closure, candidate keys, BCNF/3NF test
- ☐ B+ tree insertion; cuboid count